CBCS SCHEME

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BCS303

Third Semester B.E./B.Tech. Degree Supplementary Examination, June/July 2024

Operating Systems

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define system call. List the types of system calls.	06	L1	CO1
	b.	With a neat figure, explain the concept of virtual machines.	06	L2	CO1
	c.	Define Operating System. Explain multiprogramming and time sharing operating system.	08	L2	CO1
		OR			
Q.2	a.	List the responsibilities of the operating system for process management and memory management.	06	L1	CO1
	b.	Different between kernel mode and user mode operation of operating systems.	06	L4	CO1
	c.	Discuss the services that are provided by the operating systems for users and its efficient operation.	08	L2	CO1
		Module – 2			
Q.3	a.	Explain the process states with a neat figure.	06	L2	CO2
	b.	Differentiate between the different types of multithreading models.	06	L4	CO2
	c.	Consider the following four processes, with the length of the CPU burst given in milliseconds: Process | Arrival Time | Burst Time P1 | 0 | 8 P2 | 1 | 4 P3 | 2 | 9 P4 | 3 | 5 Computer the average waiting for the above processes using FCFS, Preemptive SJF and non-preemptive SJF scheduling algorithms.	08	L3	CO2
		OR			
Q.4	a.	Define thread. List and explain the benefits of multithreaded programming.	06	L2	CO2
	b.	Differentiate between shared memory and message passing methods for interprocess communication.	06	L4	CO2
	c.	Consider the following set of processes, with the length of the CPU-burst time given in milliseconds: Process | Arrival Time | Burst Time P_1 | 10 | 3 P_2 | 1 | 1 P_3 | 2 | 3 P_4 | 1 | 4 P_5 | 5 | 2 The processes are assumed to have arrived in the under P_1 , P_2 , P_3 , P_4 , P_5 all at time 0. Compute the average waiting time for the above processes using FCFS, Priority (Smaller priority number implies higher priority) and RR (Time quantum = 1) scheduling algorithms.	08	L3	CO2

		Module – 3			
Q.5	a.	What is critical section problem? Discuss the three requirements that a solution to critical section problem must satisfy.	06	L2	CO3
	b.	Define semaphore. Explain how mutual exclusion can be implemented using semaphores.	06	L2	CO3
	c.	Consider a system with five processes P_0 through P_4 and three resource type A has ten instances, resource type B was five instances and resource type C has seven instances. Suppose at time 70, the following snapshot of the system has been taken. Determine whether the following system is safe using Banker's algorithm. Write the safe sequence. Allocation: P_0 0 1 0; P_1 2 0 0; P_2 3 0 2; P_3 2 1 1; P_4 0 0 2 Max: P_0 7 5 3; P_1 3 2 2; P_2 9 0 2; P_3 2 2 2; P_4 4 3 3 Available: A B C = 3 3 2	08	L3	CO3
		OR			
Q.6	a.	What is deadlock? What are the necessary conditions for a deadlock to occur?	06	L2	CO3
	b.	Illustrate how dining philosophers' problem can be solved using semaphores.	06	L2	CO3
	c.	Give five memory partitions of 100 KB, 500 KB, 200 KB, 300 KB and 600 KB (in order). How would each of the first fit, best fit and worst-fit algorithms place processes of 212 KB, 417KB, 112 KB and 426 KB (in order)? Which algorithm makes the most efficient use of memory?	08	L3	CO3
		Module – 4			
Q.7	a.	Explain segmentation with an example.	06	L2	CO4
	b.	Describe the steps in handling a page fault with a neat figure.	06	L2	CO4
	c.	Consider the following reference string 7, 0, 1, 20, 3, 0, 42, 3, 03, 2,1, 2, 0, 1, 7, 0, 1 How many page faults would occur for the following replacement algorithms assuming three frames? (i) FIFO page replacement (ii) LRU page replacement	08	L3	CO4
		OR			
Q.8	a.	What is thrashing? How can it be controlled?	06	L2	CO4
	b.	Compare and contrast internal and external fragmentation of memory.	06	L2	CO4
	c.	Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6 How many page faults would occur for the following replacement algorithms assuming four frames. i) LRU replacement ii) Optimal replacement	08	L3	CO4

		Module – 5			
Q.9	a.	Compare and contrast sequential access and direct access methods for extracting information from files.	06	L4	CO5
	b.	Describe the concept of protection domain with an example of a system with three protection domains.	06	L2	CO6
	c.	Suppose that a disk drive has 200 cylinders, numbered 0 to 199. The drive is currently serving a request at cylinder 53, and the previous request was at cylinder 20. The queue of pending requests in FIFO order is 98, 183, 37, 122, 14, 124, 65, 67 Starting from current head position, what is a total distance (in cylinders) that the disk arm move to satisfy all pending requests, for each of the following scheduling algorithms? i) SSTF ii) C-SCAN	08	L3	CO5
		OR			
Q.10	a.	Differentiate between acyclic-graph directories and tree structured directories.	06	L4	CO5
	b.	Illustrate the concepts of access matrix with suitable examples.	06	L2	CO6
	c.	Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and suppose the previous request was at cylinder 125. The queue of pending requests in FIFO order is 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from current head position, what is the total distance (in cylinders) that the disk arm move to satisfy all the pending requests, for each of the following disk scheduling algorithms: i) FCFS ii) SCAN	08	L3	CO5

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CBCS SCHEME

BCS3[illegible]

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Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Operating Systems

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define Operating System. Explain dual mode of OS with a neat diagram.	5	L1, L2	CO1
	b.	Distinguish between the following terms: i) Multiprogramming and Multitasking ii) Multiprocessor system and clustered system.	10	L2	CO1
	c.	With a neat diagram, explain the concept the concept of VM-WARE architecture.	5	L1, L2	CO1
		OR			
Q.2	a.	Explain the operating system services with respect to programs and users.	5	L2	CO1
	b.	List and explain the different computing environments.	5	L1, L2	CO1
	c.	What are system calls? List and explain the different types of system calls.	10	L1, L2	CO1
		Module – 2			
Q.3	a.	Define process. Explain different states of a process with state diagram.	8	L1, L2	CO1
	b.	What is IPC? Explain direct and indirect communication with respect to message passing.	8	L1, L2	CO2
	c.	Explain context-switching.	4	L2	CO2
		OR			
Q.4	a.	What is multi-threaded process? Explain the four benefits of multithreaded programming.	6	L2	CO2
	b.	Calculate the average waiting time and average turn around time by drawing the Gantt-chart using FCFS, SJF-non preemptive, SRTF, RR(q = 2ms) and porosity algorithms.	14	L3	CO2
		Module – 3			
Q.5	a.	What is critical section? What are the requirements for the solution to critical section problem? Explain Peaterson's solution.	8	L1, L2	CO3
	b.	Explain Reader's-Writer's problem using semaphores.	12	L2	CO3

Process	Arrival time	Burst time	Porosity
P1	0	9	3
P2	1	4	2
P3	2	9	1
P4	3	5	4

OR

Q.6	a.	What is deadlock? What are the necessary conditions for the deadlock to occur?	6	L1, L2	CO3
	b.	Consider the following snap-shot of a system: Answer the following using Banker's algorithm: i) Is the system in safe state? If so give the safe sequence. ii) If process P2 requests (0, 1, 1, 3) resource can it be granted immediately.	14	L3	CO3

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	2	0	0	1	4	2	1	2	3	3	2	1
P1	3	1	2	1	5	2	5	2				
P2	2	1	0	3	2	3	1	6				
P3	1	3	1	2	1	4	2	4				
P4	1	4	3	2	3	6	6	5				

Module – 4

Q.7	a.	What is paging? Explain with neat diagram paging hardware with TLB?	10	L1, L2	CO4
	b.	What are the commonly used strategies to select a free hole from the available holes?	6	L1	CO4
	c.	Explain fragmentation in detail.	4	L2	CO4

OR

Q.8	a.	With a neat diagram? Describe the steps in handling the page fault.	8	L2	CO4
	b.	Consider the page reference string: 1, 0, 7, 1, 0, 2, 1, 2, 3, 0, 3, 2, 4, 0, 3, 6, 2, 1 for a memory with 3 frames. Determine the number of page faults using F1, F0, optimal and LRU replacement algorithms which algorithm is more efficient.	12	L3	CO4

Module – 5

Q.9	a.	Define file. List and explain the different file attributes and operations.	10	L1	CO5
	b.	Explain the different allocation methods.	10	L2	CO5

OR

Q.10	a.	What is Access Matrix? Explain Access Matrix method of system protection with domain as objects and its implementation.	10	L1, L2	CO5
	b.	A drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request 143 and previously serviced a request at 125. The queue of pending requests in FIFO order is: 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 starting from current head position. What is the total distance travelled (in cylinders) by disk arm to satisfy the requests using FCFS, SSTF, SCAN, LOOK and C-LOOK algorithm.	10	L3	CO5

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Model Question Paper-I with effect from 2023-24 (CBCS Scheme)

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Third Semester B.E. Degree Examination
Operating Systems

Max. Marks: 100

TIME: 03 Hours

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE.**

			*Bloom's Taxonomy Level	Marks
		Module -1		
Q.01	a	Distinguish between the following terms. (i) Multiprogramming and Multitasking (ii) Multiprocessor System and Clustered System	L2	6
	b	Define operating Systems. Explain the dual-mode operating system with a neat diagram.	L2	7
	c	With a neat diagram, explain the concept of the virtual machine.	L2	7
		OR		
Q.02	a	Explain the layered approach of operating system structure with a supporting diagram.	L2	6
	b	What are system calls? Briefly point out its types with illustrations.	L2	6
	c	Explain the services of the operating system that are helpful for the user and the system.	L2	8
		Module-2		
Q. 03	a	With a neat diagram, explain the states of a process with a transition diagram and process control block.	L2	6
	b	What is inter-process communication? Discuss message passing and the shared memory concept of IPC.	L2	6
	c	Calculate average waiting and turnaround times by drawing the Gantt chart using FCFS and RR (q=2ms). Processes / Arrival Time / Burst Time P1 / 0 / 9 P2 / 1 / 4 P3 / 2 / 9 P4 / 3 / 5	L3	8
		OR		
Q.04	a	Discuss in detail the multithreading model, its advantages and disadvantages with suitable illustration.	L2	7
	b	Explain five different scheduling criteria used in the computing scheduling mechanism.	L2	5
	c	Calculate the average waiting time and the average turnaround time by drawing the Gantt chart using SRTF and the Priority scheduling algorithm. Processes / Arrival Time / Burst Time / Priority P1 / 0 / 8 / 3 P2 / 1 / 4 / 2 P3 / 2 / 9 / 1 P4 / 3 / 5 / 4	L3	8

<table>
<tr><td colspan="2"></td><td>Module-3</td><td></td><td></td></tr>
<tr><td>Q. 05</td><td>a</td><td>Define deadlock. What are the necessary conditions for deadlock to occur?</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>b</td><td>Illustrate Peterson's solution for the critical section problem.</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>c</td><td>Consider the following snapshot of the system:
<table><tr><td></td><td colspan="4">Allocation</td><td colspan="4">Max</td><td colspan="4">Available</td></tr><tr><td></td><td>A</td><td>B</td><td>C</td><td>D</td><td>A</td><td>B</td><td>C</td><td>D</td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>P0</td><td>0</td><td>0</td><td>1</td><td>2</td><td>0</td><td>0</td><td>1</td><td>2</td><td>1</td><td>5</td><td>2</td><td>0</td></tr><tr><td>P1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>7</td><td>5</td><td>0</td><td></td><td></td><td></td><td></td></tr><tr><td>P2</td><td>1</td><td>3</td><td>5</td><td>4</td><td>2</td><td>3</td><td>5</td><td>6</td><td></td><td></td><td></td><td></td></tr><tr><td>P3</td><td>0</td><td>6</td><td>3</td><td>2</td><td>0</td><td>6</td><td>5</td><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>P4</td><td>0</td><td>0</td><td>1</td><td>4</td><td>0</td><td>6</td><td>5</td><td>6</td><td></td><td></td><td></td><td></td></tr></table>Answer the following questions using Banker's algorithm:
a. What is the content of the need matrix?
b. Is the system in a safe state? If yes, mention the safe sequence.
If a request from process P1 arrives for (0,4,2,0) can the request be granted immediately?</td><td>L3</td><td>9</td></tr>
<tr><td colspan="2"></td><td>OR</td><td></td><td></td></tr>
<tr><td>Q. 06</td><td>a</td><td>Explain different methods to recover from deadlocks.</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>b</td><td>What is a resource allocation graph? Consider an example to explain how it is very useful in describing a deadly embrace.</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>c</td><td>What is a semaphore? State a Dining Philosopher problem gives a solution using semaphore.</td><td>L2</td><td>8</td></tr>
<tr><td colspan="2"></td><td>Module-4</td><td></td><td></td></tr>
<tr><td>Q. 07</td><td>a</td><td>What is TLB? Explain TLB in detail with a paging system with a neat diagram.</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>b</td><td>With the help of a neat diagram, explain the various steps of address binding.</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>c</td><td>Consider the page reference string: 1,0,7,1,0,2,1,2,3,0,3,2,4,0,3,6,2,1 for a memory with three frames. Determine the number of page faults using the FIFO, Optimal, and LRU replacement algorithms. Which algorithm is most efficient?</td><td>L3</td><td>8</td></tr>
<tr><td colspan="2"></td><td>OR</td><td></td><td></td></tr>
<tr><td>Q. 08</td><td>a</td><td>What is demand paging? Explain the steps in handling page faults using the appropriate diagram.</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>b</td><td>What is segmentation? Explain the basic method of segmentation with an example.</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>c</td><td>Discuss the structure of the page table with a suitable diagram.</td><td>L2</td><td>8</td></tr>
<tr><td colspan="2"></td><td>Module-5</td><td></td><td></td></tr>
<tr><td>Q. 09</td><td>a</td><td>What is a file? What are its attributes? Explain file operations.</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>b</td><td>Explain in detail about various file operations in a file system.</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>c</td><td>Discuss various directory structures with neat diagrams.</td><td>L2</td><td>8</td></tr>
<tr><td colspan="2"></td><td>OR</td><td></td><td></td></tr>
<tr><td>Q. 10</td><td>a</td><td>Explain contiguous and linked disk space allocation methods.</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>b</td><td>Explain the access matrix method of system protection with the domain as objects and its implementation.</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>c</td><td>Given the following sequences 95,180,34,119,11,123,62,64 with the track 50 and ending track 199. What is the total disk travelled by the disk arm using FCFS, SSTF, LOOK and CLOOK algorithm</td><td>L3</td><td>8</td></tr>
</table>

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of question.

CBCS SCHEME

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BCS303

Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Operating Systems

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

			M	L	C
		Module – 1			
Q.1	a.	Define Operating System. Explain dual mode of operating systems with a neat diagram.	06	L1 L2	CO1
	b.	Distinguish between the following terms: i) Multiprogramming and Multitasking ii) Multiprocessor and Clustered system	06	L2	CO1
	c.	Explain with a neat diagram VM-WARE Architecture.	08	L1 L2	CO1
		OR			
Q.2	a.	List and explain the services provided by OS for the user and efficient operation of system.	06	L2	CO1
	b.	Explain the different computing equipments.	06	L2	CO1
	c.	What are systems calls? List and explain the different types of systems calls.	08	L1 L2	CO1
		Module – 2			
Q.3	a.	What is process? Explain process state diagram and process control block with a neat diagram.	10	L1 L2	CO2
	b.	What is interprocess communication? Explain direct and indirect communication with respect to message passing system.	10	L1 L2	CO2
		OR			
Q.4	a.	List and explain the different types of multithreading models.	06	L1 L2	CO2
	b.	Calculate the average waiting time and average turnaround time by drawing the Gantt-chart using FCFS, SJF, RR (Q = 4ms) and priority scheduling (Higher Number is having highest priority).	14	L3	CO2
		Module – 3			
Q.5	a.	What is critical section? Give the Peterson's solution to 2 processes critical section problem.	05	L1 L2	CO3
	b.	Explain Reader's and Writer's problem in detail.	07	L2	CO3
	c.	What is semaphore? Discuss the solution to the classical dinning philosopher problem.	08	L1 L2	CO3

Process	B.T. (ms)	Priority
P_1	24	1
P_2	03	2
P_3	03	3

OR

Q.6	a.	What is a Deadlock? What are the necessary conditions for the deadlock to occur?	06	L1 L2	CO3
	b.	Consider the following snap shot of the system. Answer the following questions: i) What is the content of the matrix need? ii) Is the system on a safe state? If so, find safe sequence. iii) If P_1 requirements for (1, 0, 2) additional resources can P_1 be granted.	14	L3	CO2

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P_0	0	1	0	7	5	3	3	3	2
P_1	2	0	0	3	2	2			
P_2	3	0	2	9	0	2			
P_3	2	1	1	2	2	2			
P_4	0	0	2	4	3	3			

Module – 4

Q.7	a.	What is paging? Explain with a neat diagram paging hardware with TLB.	10	L1 L2	CO4
	b.	Explain the different strategies used to select a free hole from available holes.	05	L1	CO4
	c.	What is Fragmentation? List and explain its types.	05	L2	CO4

OR

Q.8	a.	What is page fault? With a neat diagram explain steps in handling page fault.	08	L2	CO4
	b.	Consider the page reference string for a memory with 3 frames determine the number of page faults using FIFO, optimal and LRU replacement algorithms. Which algorithms is more efficient? 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1	12	L3	CO4

Module – 5

Q.9	a.	Define File. List and explain different file operations and file attributes.	10	L1	CO5
	b.	Explain the different file allocation methods.	10	L2	CO5

OR

Q.10	a.	What is Access Matrix? Explain the implementation of Access Matrix.	10	L2	CO5
	b.	A drive has 5000 cylinders numbered 0 to 4999. The drive is currently servicing at a request 143 and previously served a request at 125. The queue of pending request in FIFO order. 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 starting from current head position. What is the total distance travelled (in cylinders) by a disk arm to satisfy the request using FCFS, SSTF, SCAN, LOOK and C-Look algorithm	10	L3	CO5

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BCS303

Third Semester B.E./B.Tech. Degree Examination, June/July 2024
Operating Systems

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain multi programming and time sharing systems.	07	L3	CO1
	b.	Explain the dual mode operation in operating systems with a neat block diagram.	07	L3	CO1
	c.	What are virtual machines? Explain with a neat figure.	06	L3	CO1
		OR			
Q.2	a.	What are system calls? Briefly explain different types of system calls.	07	L3	CO1
	b.	List and explain the services provided by OS for the user in efficient operation of a system.	07	L3	CO1
	c.	What are micro kernels? With a neat figure, explain the micro kernel structure? Point out their advantages over layered approach.	06	L3	CO1
		Module – 2			
Q.3	a.	What is process? Explain different states of the process with state transition diagram and process control block.	08	L2	CO2
	b.	What is Interprocess communication? Explain.	06	L2	CO2
	c.	What is thread? How it is different from process? Discuss various multithreading models with suitable illustration.	06	L2	CO2
		OR			
Q.4	a.	Consider the following processes where smaller the number has higher priority. Draw the Gantt chart compute the waiting time and average turnaround time by using FCFS, SRTF, preemptive priority scheduling.	12	L2	CO2
	b.	Discuss the benefits of multithreaded programming. Explain the threading issues in detail.	08	L2	CO2
		Module – 3			
Q.5	a.	What are the requirements that must be satisfied by a solution to the critical section problem? Illustrate with an example the Peterson's solution for critical section problem.	08	L3	CO3
	b.	What is critical section problem and solutions to the problem? How to solve using semaphores?	06	L3	CO3
	c.	Explain the classical bounded buffer problem of synchronization. Give the solution	06	L3	CO3
		OR			
Q.6	a.	What is dead lock? What are the necessary conditions for the deadlock to occur? How to recover from deadlocks.	10	L3	CO3

Processes	Arrival times	Burst time	Priority
P_1	0	7	4
P_2	3	5	2
P_3	3	3	6
P_4	5	5	3

	b.	Assume that there are 5 processes P_0 to P_4 and 4 types of resources. At time T_0 the system has following: Apply the bankers algorithm to answer following: (i) What is the content of need matrix? (ii) Is the system in a safe state? (iii) If the request from P_1(2, 1, 1, 0) arrives can it be granted?	10	L3	CO3

Processes	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P_0	0	1	1	0	0	2	1	0	1	3	1	0
P_1	1	4	4	1	1	6	5	2				
P_2	1	3	6	5	2	3	6	6				
P_3	0	6	3	2	0	6	5	2				
P_4	0	0	1	4	0	6	5	6				

Module – 4

Q.7	a.	What is paging? Differentiate between paging and segmentation.	06	L3	CO4
	b.	What are TLB? Explain TLB in detail with a simple paging system and neat diagram.	08	L3	CO4
	c.	Given the memory partitions of 100K, 500K, 200K, 300K and 600K, apply first fit, best fit and worst fit algorithms to place 212K, 417K, 112K and 426K.	06	L3	CO4

OR

Q.8	a.	What is page fault? With a neat diagram, explain the steps in handling page fault.	08	L3	CO4
	b.	Illustrate how demand paging affects system performance. What is thrashing how it can be controlled?	06	L3	CO4
	c.	Consider the following sequence: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 Assuming frame size of 4, apply LRU, FIFO and optimal algorithm to find the page faults. Find out which algorithm is most efficiency.	06	L3	CO4

Module – 5

Q.9	a.	Explain various file attributes and operations of files.	06	L3	CO5 CO6
	b.	With a neat diagram, explain two level and tree structured directory structure.	08	L3	CO5 CO6
	c.	What is file? Explain the file mounting.	06	L3	CO5 CO6

OR

Q.10	a.	Give the following sequence: 95, 180, 34, 119, 11, 123, 62, 64 with the head initially at 50 and ending at track 199. What is the total disk travelled by the disk drum to satisfy request using FCFS, SSTF, LOOK and CLOOK algorithms.	12	L3	CO5 CO6
	b.	Explain the access matrix model of implementing protection on OS.	08	L3	CO5 CO6

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Fourth Semester B.E. Degree Examination, July/August 2022
Operating Systems

Time: 3 hrs. Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1 a. Distinguish between the following terms:
i) Multi programming and multitasking.
ii) Multi processor systems and clustered systems. **(10 Marks)**

b. Define Operating Systems. Explain dual mode of operating systems with a neat diagram. **(05 Marks)**

c. Explain about system calls with an example of handling a user application invoking the open( ) system call. **(05 Marks)**

OR

2 a. What is a process? Illustrate with a neat diagram the different states of a process and control block. **(05 Marks)**

b. Discuss the implementation of IPC using message passing systems in detail. **(10 Marks)**

c. List and explain the services provided by OS for the user and efficient operation of system. **(05 Marks)**

Module-2

3 a. Give a brief description about multithreading and explain the different multi threading models. **(05 Marks)**

b. Discuss the issues that come with multithreaded programming. **(10 Marks)**

c. Explain CPU scheduling criteria. **(05 Marks)**

OR

4 a. Calculate the average waiting time and the average turnaround time by drawing the Gantt chart using FCFS, SRTF, RR (q = 2ms) and priority algorithms. Lower priority number represents higher priority.

Process	Arrival Time	Burst Time	Priority
P_1	0	9	3
P_2	1	4	2
P_3	2	9	1
P_4	3	5	4

(12 Marks)

b. What is critical section problem? What are the requirements for the solution to critical section problem? Explain Peterson's solution. **(08 Marks)**

Module-3

a. What is a deadlock? What are the necessary conditions for the deadlock to occur? **(05 Marks)**
b. How to prevent the occurrence of deadlock, explain in detail. **(05 Marks)**
c. Consider the following snapshot of a system:

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P_0	2	0	0	1	4	2	1	2	3	3	2	1
P_1	3	1	2	1	5	2	5	2				
P_2	2	1	0	3	2	3	1	6				
P_3	1	3	1	2	1	4	2	4				
P_4	1	4	3	2	3	6	6	5				

Answer the following using Banker's algorithm.
i) Is the system in safe state? If so, give the safe sequence.
ii) If process P_2 requests (0, 1, 1, 3) resources can it be granted immediately? **(10 Marks)**

OR

a. Explain paging hardware with TLB. **(05 Marks)**
b. Explain segmentation in detail. **(05 Marks)**
c. Discuss structure of page table with suitable diagrams. **(10 Marks)**

Module-4

a. Describe the steps in handling page faults. **(06 Marks)**
b. Consider the page reference string: 1, 0, 7, 1, 0, 2, 1, 2, 3, 0, 3, 2, 4, 0, 3, 6, 2, 1 for a memory with 3 frames. Determine the number of page faults using FIFO, optimal and LRU replacement algorithms. Which algorithm is most efficient? **(14 Marks)**

OR

a. Explain the different allocation methods. **(10 Marks)**
b. Discuss the various directory structures with required diagrams. **(10 Marks)**

Module-5

a. Explain access matrix method of system protection with domain as objects and its implementation. **(10 Marks)**
b. A drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request at 143 and previously serviced a request at 125. The queue of pending requests in FIFO order is: 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from current head position, what is the total distance travelled (in cylinders) by disk arm to satisfy the requests using FCFS, SSTF, SCAN, LOOK and C-LOOK algorithms. **(10 Marks)**

OR

a. With a neat diagram, explain the components of a Linux system. **(08 Marks)**
b. Explain the different IPC mechanisms available in Linux. **(06 Marks)**
c. Discuss about scheduling in Linux. **(06 Marks)**

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C.M.R. INSTITUTE OF TECHNOLOGY CMR BANGALORE

Fourth Semester B.E. Degree Examination, June/July 2024
Operating Systems

Time: 3 hrs. Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

Module-1

1 a. Explain what is an operating systems. Discuss the role of operating system from different view points. **(06 Marks)**

b. Explain the dual mode of operation of an operating system. **(07 Marks)**

c. List and explain the different states of process with a neat diagram. **(07 Marks)**

OR

2 a. With a neat diagram, explain the concept of virtual machine. **(07 Marks)**

b. What are system calls? Explain the handling of open ( ) system call. **(07 Marks)**

c. List the different operating system services and explain. **(06 Marks)**

Module-2

3 a. Explain the different multi threading models with a diagram. List out the different threading issues with multi threaded models. Explain any one. **(10 Marks)**

b. Calculate the average waiting time by drawing Gantt chart using first come first serve, shortest job first and round robin scheduling algorithms for the following process. Time quantum = 4

Process	Burst time
P_0	24
P_1	3
P_2	3

(10 Marks)

OR

4 a. What is a critical section problem? List and explain the solution to this problem using Peterson's algorithm. **(10 Marks)**

b. What is Bounded Buffer problem? Explain how semaphores can be used for synchronization in this problem. **(10 Marks)**

Module-3

5 a. What is a deadlock? Explain how can it be prevented. **(10 Marks)**

b. Explain Banker's algorithm for deadlock avoidance. **(10 Marks)**

OR

6 a. Given a system with total resources A(3) ; B(14) and C(12), determine whether the following system is in safe state or not using Banker's algorithm.

	Allocation			Max			Availability		
	A	B	C	A	B	C	A	B	C
P_0	0	0	1	0	0	1	1	5	2
P_1	1	0	0	1	7	5			
P_2	1	3	5	2	3	5			
P_3	0	6	3	0	6	5			
P_4	0	0	1	0	6	5			

Justify can $P_1(0, 0, 1)$ request be granted immediately? **(10 Marks)**

b. Explain the concept of paging with a neat diagram. **(10 Marks)**

Module-4

What is demand paging? Explain the steps in handling a page fault with a neat diagram.

Consider the following page reference stream, (10 M

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

How many page faults would occur for LRU and FIFO page replacement algor assuming 3 Frames? What is Belady's anomaly? (10 M

OR

Explain the two structures used to implement file systems.

List the different methods of disk allocation and explain any one. (10 M

List the different free space management approaches and explain any two. (05 M (05 M

Module-5

Explain access matrix and list its implementation.

Explain various disk scheduling algorithms with an example. (10 M (10 M

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OR

Explain the various components of the Linux system.

Explain disk formatting, boot block and bad block. (10 M (10 M

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18CS43

Fourth Semester B.E. Degree Examination, Jan./Feb. 2023
Operating Systems

Time: 3 hrs Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

Module-1

1 a. Define Operating System. Explain briefly multiprogramming and time sharing system. **(08 Marks)**

b. Explain various OS services. **(05 Marks)**

c. What are system calls? Briefly explain types of system call. **(07 Marks)**

OR

2 a. With a neat diagram, explain the different state of process. **(07 Marks)**

b. Explain scheduler in a process execution. **(05 Marks)**

c. Explain direct and indirect communication with respect to message passing systems. **(08 Marks)**

Module-2

3 a. Discuss multithreading models. **(05 Marks)**

b. Discuss the benefits of multithreaded programming. **(05 Marks)**

c. Consider the following set of four process with length of CPU burst given in MS:

Process	Burst time
P1	24
P2	3
P3	3

Compute the waiting time and avg. turnaround time for the above process using FCFS scheduling algorithm. **(10 Marks)**

OR

4 a. Explain requirements must satisfy to critical section problem. **(04 Marks)**

b. Illustrate with an example Peterson's solution problem. **(08 Marks)**

c. Explain syntax and schematic view of monitors. **(08 Marks)**

Module-3

5 a. What are necessary conditions for deadlock? **(04 Marks)**

b. Explain different methods to recovery from deadlocks. **(08 Marks)**

c. Consider the following snapshot of system:

Process	Allocation			Maximum			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

Find the need matrix and calculate safe sequence using bankers algorithm-mention the above is safe or not safe. **(08 Marks)**

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OR

6 a. What is paging hardware with TLB?
b. Explain structure of page table with respect to hierarchical paging.
c. Explain the process of segmentation.

Module-4

7 a. Describe the steps in handling a page fault.
b. Explain copy on write process in virtual machine.
c. Explain FIFO and optimal page replacement algorithm.

OR

8 a. Explain with a diagram any two disk allocation method in detail.
b. List the different operations performed on a directory.
c. Explain tree structured directory structure.

Module-5

9 a. Explain SCAN, CSCAN and LOOK scheduling techniques.
b. Explain access matrix model of implementing protection in OS.
c. Explain bad-block recovery in detail.

OR

10 a. With a diagram, explain the component of LINUX system.
b. Explain the IPC mechanisms in LINUX.
c. Discuss passing of data among process in LINUX system.

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